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Mobile Computing

CO1-AT2 · SET - I

Class Test - 1

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47/50

1) 45MHZ ⇒ 45000

So, $\frac{45000}{50} = 900$.

2) 200 KHZ

So, 100 carriers ; each 8 time.

⇒ N = 100 x 8 ⇒ 800 voice channels.

3) carrier-to-interference (C/I)

C/I = -6 - (-92).

4) path-loss exponent n=4

$Pr = \frac{64}{2^4} = \frac{64}{16}$

= 4w

5) Total channels in a cluster

N = 4 x 15

= 60 channels.

6) TDMA.

$T = \frac{5ms}{8} = 0.625$ ms.

7) power in dBm.

Ew to dBm :-

$P_t = 10 \log_{10}(8000) + 30 = 39.03$ dBm.

$Pr = 39.03 - 33$

⇒ 6.03 dBm.

8) Cluster call capacity :-

N = 6 x 250

= 1500 simultaneous calls.

17) Total bandwidth consumed.

$B = 150 \times 25\,KHz$.

$= 3750\ KHz$

$= 3.75\ KHz$.

18) 90 KM/h.

1.5 KM radius:-

$90/15 = 6$. Call diameter $= 2 \times 1.5 = 3$.

$N = 90/3$

$= 30$ handoffs.

21) No. of cells required:-

$R = 400m = 0.4Km$.

$A = (0.4)^2 = 0.16Km^2$

$N = \frac{400}{0.16} = 2500$ cells

23) offered traffic in Erlangs:-

$A = \frac{\text{subscribers} \times \text{calls/hour} \times \text{call duration in min}}{60}$

$= \frac{1200 \times 4 \times 2.5}{60}$

$= 200$ Erlangs.

25) total network user capacity:-

$N = 1200 \times 600$.

$= 720,000$ users.

27) offered traffic

$A = 40,000 \times 3 = 120,000$

120,000 − 100,000

= 20,000.

29) $U = \frac{9000}{12000} \times 100$ 2

= 75%.

31) $N = \frac{30\ MHz}{200\ MHz}$

$\frac{3000}{20} = \frac{3000}{20}$ 2

= 150 channels.

33) $P_t = 10\log_{10}(12000) + 30$

$= 40.79\ dBm$

$= 40.79 - 45$ 2

$= -4.21\ dBm$.

35) TDMA slot

$T = \frac{4.8\ m.s}{8}$ 2

$= 0.6\ ms$

37) % offered calls blocked

$A = 70,000 \times 4$

$= 280,000$ 2

280000 − 250,000

= 30,000

Blocked (%) $= \frac{30,000}{280,000} \times 100$

$= 10.71\%$.

39) $N = \frac{0.4}{100} \times 60{,}000$

$= 240$ calls per second.

41) $A = 2.5 \times 150$

$= 375\ km^2$.

43) No. of paging cycles.

$N = \frac{150{,}000}{6000}$

$= 25$ cycles.

45) (SNR)

$SNR = P_S - P_n$.

$SNR = -85 - (-115)$

$= 30\ dB$.

47) $A = \frac{1500 \times 4 \times 2.5}{60}$

$= 250$ triangles.

49) $T = 1500 \times 600\ Kbps$

$T = 900{,}000\ Kbps$

$= 900\ Mbps$.